Web Development

HTML

❖ tag

- Div tag is a block level element takes up full width available.
- Creates line breaks before and after. Used for grouping larger section of content.
- Creates structural divisions in a webpage.

❖ tag

- Span tag is a inline element only takes up necessary width.
- No line breaks before or after. Used for grouping smaller pieces of content.
- Best for styling specific text portions.

Example –

```
<body>
 <div style="background-color:red;">
 <p> Hello World !</p>
 </div>
 <p> Hello India </p>
 <span style="background-color:red;">
  <a>hello hira</a>
 </span>
</body>
```

HIRA KUMAR

```html
<body>
  <div style="background-color: red;">
  <p> Hello World !</p>
  </div>
  <p> Hello India </p>
  <span style="background-color: red;">
    <a>hello hira</a>
  </span>
</body>
</html>
```

http://127.0.0.1:5500/index.html 260%

Hello World !

Hello India

hello hira

❖ Difference <p> tag & <a> tag

<p> tag	<a> tag
• P is called is paragraph tag • Block – level element • Takes up full width • Represents a paragraph of text • Creates line breaks before and after	• A is called anchor/link tag • Inline element • Only takes necessary width • No line breaks before or after • Creates hyperlinks to other pages, sections, or resources. • Requires href attribute to be functional.

```html
<body>
<p style ="background-color: yellow;">
  Hello India
</p>
<br>
<a style ="background-color: yellow;">
  Hello Hira
</a>
</body>
```

http://127.0.0.1:5500/index.html 260%

Hello India

Hello Hira

❖ Semantic Tags

- Semantic tags in HTML are elements that clearly describe their meaning to both the browser and the developer.
  (वैसा tag जिसे देखते ही उसका meaning पता चल जाए की उसके अंदर क्या content है |)
- Example:-

<header>, <nav>, <article>,<section>,<footer>, <image>,<video>,<p>, <a>,....

➢ Why use semantic tags ?

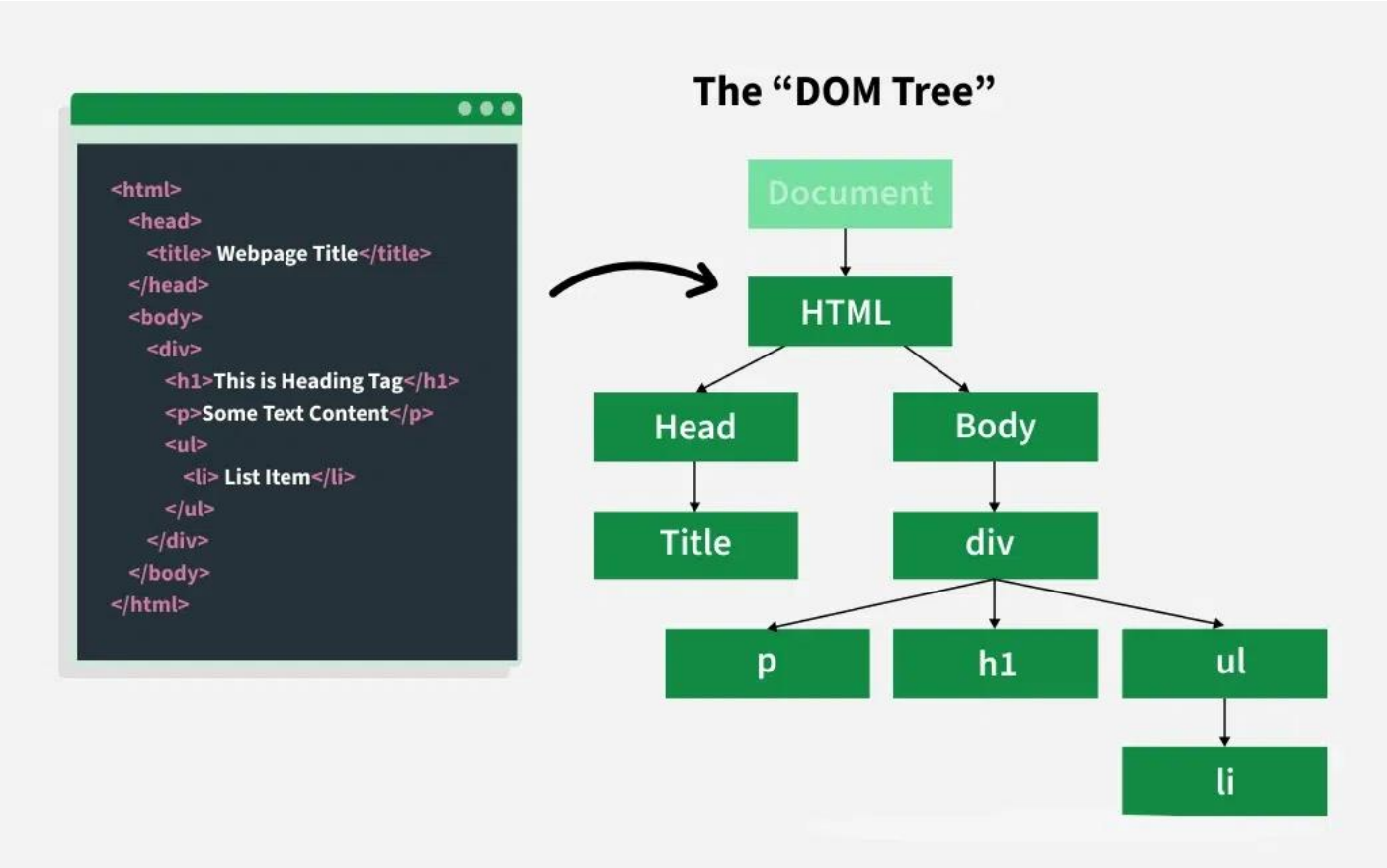
i. **Accessibility** → Screen readers can better interpret the structure.
ii. **SEO** → Search engines understand the content hierarchy.
iii. **Maintainability** → Code is easier to read and maintain.
iv. **Clarity** → Developers can quickly understand the layout.

❖ Non Semantic

- Non-semantic tags are elements that do not convey any meaning about the content they contain. They are used purely for structure,styling, or grouping purpose.
- Example :- , ,<b>,<i>,<font>, <big>,<small>, ,<tt>,..

❖ DOM

- DOM means Document Object Model
- It is a programming interface for HTML (and XML) documents. When a web page is loaded, the browser creates a tree-like structure of the page, where each HTML element becomes an object.
- This allows programming language like JAVAScript to.
- Bridge b/w Structure (HTML) and Programming Logics (JS)
- Example –



❖ <nav> tag

- <nav> tag is a semantic HTML5 element used to define a section of navigation links on a webpage. It's designed to contain major navigation blocks that link to other pages or different sections within the same page.
- Example –

```
<body>
  <nav>
    <a href="#"> home </a>  
    <a href="#"> Registration </a>  
    <a href="#"> Login </a> 
    <a href="#"> Contact </a>
  </nav>
</body>
```

http://127.0.0.1:5500/1st pro/index.html

home Registration Login Contact

❖ <embed> tag

- The <embed> tag is an HTML element used to embed external resources like plugins, multimedia content, or interactive application directly into a web page.
- (पुराने HTML में use होता था, जब video, audio,...tag नहीं थे तब)
- Example –

```html
<body>
  <h1> Embed tag</h1>
  <embed src="bca.pdf" type="application/pdf" width="600" height="400">
  <embed src="kashi.mp4" type="video/mp4" widht="600" height="400">
  <embed src="shiv.mp3" type="audio/mpeg">
</body>
```

http://127.0.0.1:5500/index.html

Embed tag

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BCA – 1st YEAR			
HONS. PAPER	SUBJECTS	THEORY MARKS	INTERNAL PRACTICAL MARKS
I	Computer Fundamentals	75	25
	Office Automation Application (MS Office)		
	Web Application – HTML, CSS & JavaScript		
II	Programming Logic and Algorithm with C Language	75	25
	Operating System with Linux and Windows		
	Software Analysis and Design (SAD)		
SUBSIDIARY PAPERS	SUBJECTS	MARKS	
I	Subsidiary I	100	
II	Subsidiary II	100	
Composition	Hindi	100	

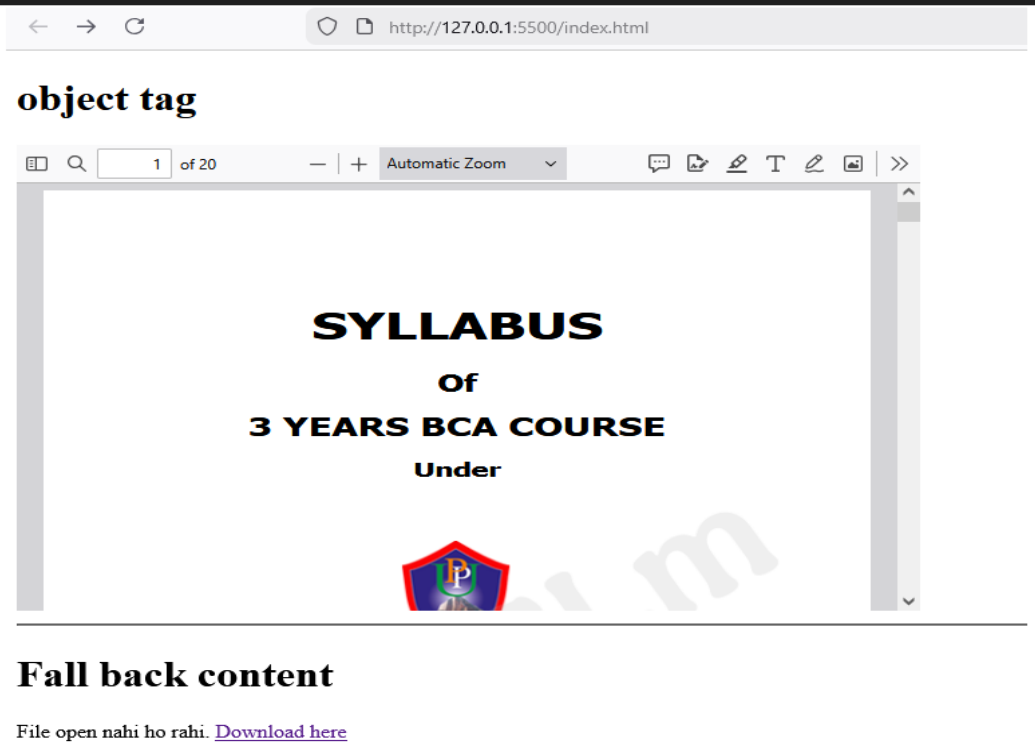
0:20 / 0:20

<object> tag

The <object> tag is a versatile HTML element used to embed external resources like images, videos, pdfs , flash, java applets and even other web pages. It’s more powerful and flexible tha <embed> because it support fallback content.

Example -

```html
<body>
  <h1> object tag</h1>
  <object data="bca.pdf" type="application/pdf" width="50%" height="400"></object> <hr>
  <h1>Fall back  content</h1>
   <object data="ca.pdf" width="400" height="300">
  <p>File open nahi ho rahi. <a href="bca.pdf">Download here</a></p>
</object>
</body>
```



❖ <time> tag

- **<time> tag is used** in HTML to represent date and time in a machine-readable format.

OR

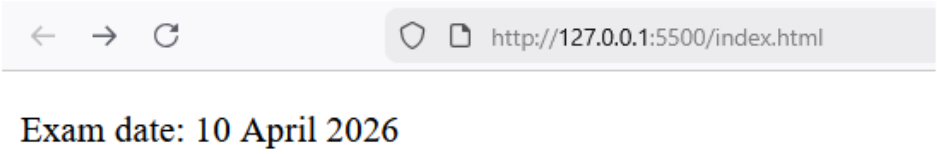
- The <time> tag is used to define a specific date or time.
- Example –

```
<p>Exam date: <time datetime="2026-04-10">10 April 2026</time></p>
```

Here-

✓ datetime attribute → machine-readable format
✓ Useful for **SEO and accessibility**.
✓ It is a **semantic tag.**

```html
<body>
  <p>Exam date: <time datetime="2026-04-10">10 April 2026</time></p>
</body>
```



Exam date: 10 April 2026

❖ XHTML

- XHTML (Extensible Hypertext Markup Language) is a stricter version of HTML and is designed to confront some of the problems associated with the different and competing versions of HTML
- XHTML is also designed to better integrate HTML with other markup languages such as XML.
- HTML will not become obsolete anytime soon.
- File extension of XHTML is .xhtml or .xht
  - You can turn an HTML document into an XHTML document by adding a few lines of code at the beginning of the page:

This must be the first item on the page

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
```

This replaces the opening <html> tag

```
<html xmlns="http://www.w3.org/1999/xhtml">
```

This is placed inside the <head> element

```
<meta http-equiv="Content-Type" content="text/html;
  charset=utf-8" />
```

Untitled - Notepad

File Edit Format View Help

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">

  <head>
    <title>Untitled Document</title>
    <meta http-equiv="Content-Type" content="text/html;
       charset=utf-8" />
  </head>
  <body>

  </body>
</html>
```

❖ Difference b/w HTML5 & XHTML

S.No.	HTML	XHTML
1.	HTML stands for Hypertext Markup Language.	XHTML stands for Extensible Hypertext Markup Language.
2.	It was developed by Tim Berners-Lee.	It was developed by W3C i.e World Wide Web Consortium.
3.	It was developed in 1991.	It was released in 2000.
4.	It is extended from SGML.	It is extended from XML and HTML.
5.	The format is a document file format.	The format is a markup language.
6.	All tags and attributes are not necessarily to be in lower or upper case.	In this, every tag and attribute should be in lower case.
7.	Doctype is not necessary to write at the top.	Doctype is very necessary to write at the top of the file.
8.	It is not necessary to close the tags in the order they are opened.	It is necessary to close the tags in the order they are opened.
9.	While using the attributes it is not necessary to mention quotes. For e.g. <Geeks>.	While using the attributes it is mandatory to mention quotes. For e.g. <Geeks="GFG">.
10.	Filename extension used are .html, .htm.	Filename extension are .xhtml, .xht, .xml.

Markup language	Percentage of Websites
HTML5	95.4%
HTML	2.9%
XHTML	0.1%